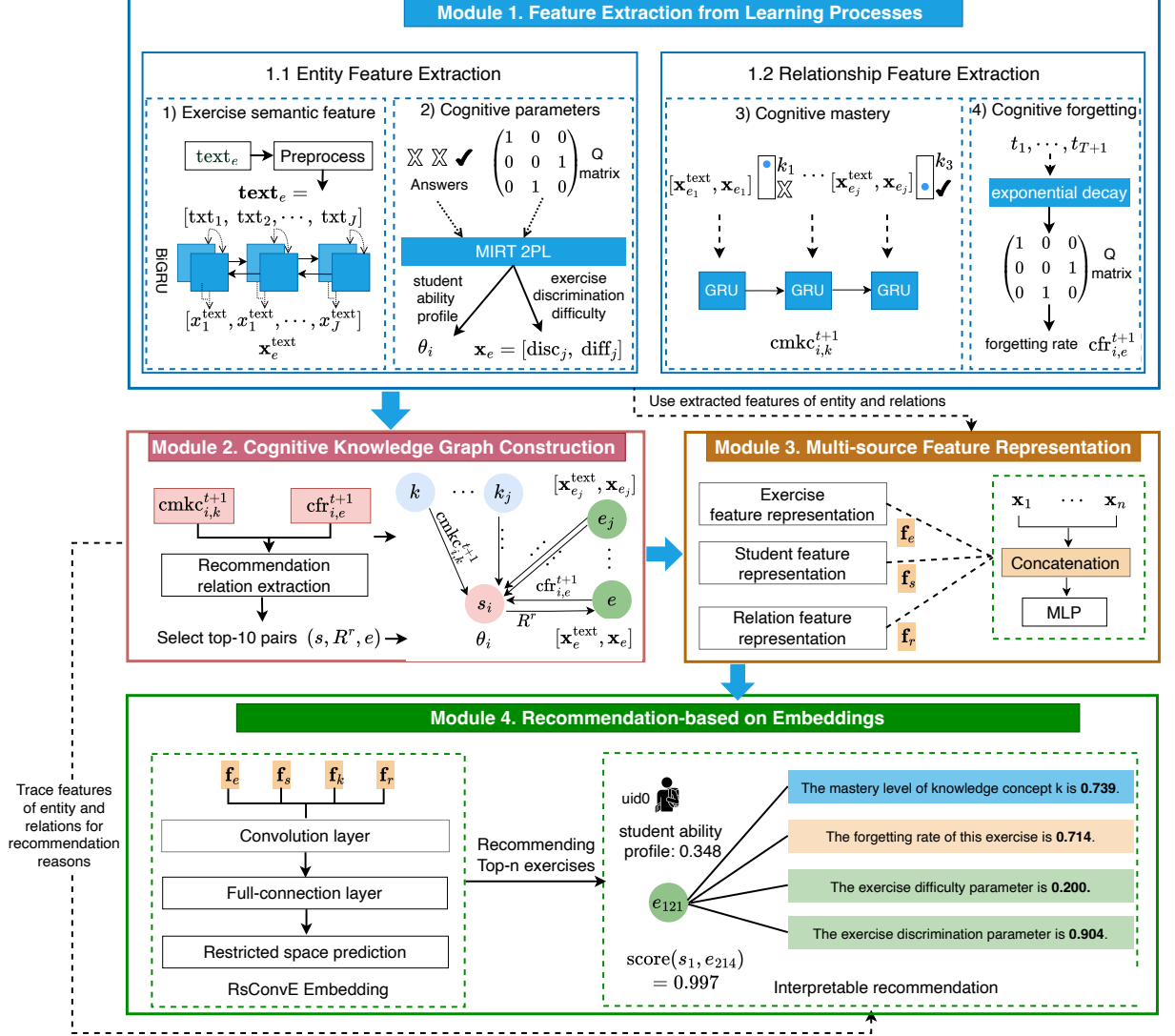


Graphical Abstract

Interpretable Personalized Exercise Recommendation via Content-Aware Cognitive Knowledge Graph Embedding

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Highlights

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- A content-aware cognitive knowledge graph (2CKG4ER) is proposed to bridge knowledge tracing and exercise recommendation.
- Textual semantics, IRT cognitive parameters, and dynamic mastery-forgetting states are integrated with a novel RsConvE model for link inference.
- Experiments on three datasets demonstrate superior recommendation adaptivity and transparent pedagogical interpretability.

Interpretable Personalized Exercise Recommendation via Content-Aware Cognitive Knowledge Graph Embedding^{*}

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ABSTRACT

Personalized exercise recommendation in online learning systems requires aligning tasks with students' evolving cognitive states while providing transparent justifications of recommendation decisions to foster trust. However, explosive resource growth and dynamic student cognitive states make accurate, interpretable recommendations difficult. Existing learning-based exercise recommendation methods fail to jointly integrate exercise semantic features into both learner representations and recommendation decisions, creating a critical gap in accurate recommendation and in explaining the reason behind recommendations. To address this issue, we propose 2CKG4ER, a method that delivers personalized and interpretable exercise recommendations via a content-aware cognitive knowledge graph. First, the 2CKG4ER extracts semantic of exercises and cognitive features of exercises and students alongside the relational features connecting students, exercises, and knowledge concepts. It then utilizes an MLP-based feature fusion module to integrate these multi-source attributes. Next, 2CKG4ER introduces RsConvE, a variant ConvE of knowledge graph embedding model, that restricts the tail prediction space. Extensive experiments on three educational datasets demonstrate that 2CKG4ER achieves competitive adaptivity, outperforming the second-best baseline by approximately 2.9%, maintains stable novelty performance, and provides interpretable recommendations for exercises.

1. Introduction

Online learning platforms provide students access to interactive learning materials without geographical constraints, making them increasingly popular in education. To enhance learning effectiveness in these platforms, personalized learning has emerged as a hot research topic, where exercises are adaptively recommended with an appropriate level of difficulties. A key challenge in personalized learning is aligning recommendations with students' evolving learning states. Knowledge tracing (Abdelrahman et al., 2023), which forms the foundation of exercise recommendation, models learning dynamics and predicts future performance using sequential models. Based on the estimated knowledge states, exercise recommendation aims to provide exercises that effectively support learning in line with students' cognitive needs.

Existing personalized exercise recommendation approaches mainly fall into two categories: collaborative filtering (CF)-based methods (Li et al., 2021; Ngo et al., 2024) and learning-based methods (Cheng et al., 2026; Yang et al., 2026; Liu et al., 2019; Xian et al., 2019). Collaborative filtering is a widely adopted technique in educational recommendation scenarios. It generates recommendations by identifying similarities between learners or exercises, showing advantages such as simplicity and ease of application even in the absence of explicit instructional knowledge structures. However, despite their efficiency, these approaches often rely on shallow behavioral data (e.g., learner response logs). They fail to capture students' deeper cognitive states or trace how those states evolve, leading to recommendations that lack strong educational relevance (Guan et al., 2025).

In contrast, learning-based recommendation approaches, such as deep reinforcement learning (Cai et al., 2019; Cheng et al., 2026), recommend items based on students' evolving knowledge states, which are inferred by knowledge tracing (KT) models (Wang et al., 2025a; Gao and Liu, 2025) from their past interaction logs. However, a disconnect

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often persists because recommendation modules and KT modules operate separately; the recommendation targets exercises, while the estimated knowledge states from KT overlook specific exercise features. This misalignment is likely to heighten the mismatch between the recommended exercises and students' evolving states. Moreover, these deep learning models typically operate as "black boxes", making their decisions difficult to interpret and reducing users' trust in the recommendation system. Although explainable recommendation systems (Yang et al., 2024) offer promising solutions in e-commerce, they fail to extract sufficient educational features from the learning process, leaving them unsuitable to provide pedagogically convincing justifications underlying their recommendations.

Recently, knowledge graphs have been increasingly integrated into personalized learning recommendation systems (Zheng et al., 2025; Fan et al., 2026; Hong et al., 2025). These knowledge graphs can capture complex relationships among students, exercises, and knowledge concepts, thus can offering interpretable recommendation clues using the graph's structure. Existing graph-based recommendation studies usually model student-resource interactions or knowledge concept associations as graph links and then use graph embedding models to predict potential recommendation relations among student and exercises. However, these methods mainly rely on ID-based embedding (Guan et al., 2023, 2025), overlooking rich semantic information and cognitive parameters of the exercises. Even for the same knowledge concept, exercises with varying semantic content, difficulty levels, and discrimination power can differ in their suitability for students with different abilities. Even when associated with the same knowledge concept, exercises with distinct semantic texts, difficulty levels, and discrimination power are suited for students with different abilities. Neglecting these attributes may lead to inaccurate recommendations and incomplete interpretability by overlooking essential exercise characteristics.

To overcome these limitations, this study formulates personalized exercise recommendation as a multi-relational cognitive knowledge graph construction and inference problem. We propose 2CKG4ER, a content-aware cognitive knowledge graph-based recommendation method. Specifically, exercise semantic features are extracted from textual descriptions using a BiGRU model, while cognitive attributes, including difficulty, discrimination, and learner overall ability, are estimated using item response theory. Learner mastery levels over knowledge concepts which are derived from knowledge tracing models and forgetting patterns which are modeled by exponential decay are incorporated as relational features. Based on both cognitive relations, recommendation relations are further constructed to capture personalized student-exercise interactions, forming the content-aware cognitive knowledge graph. The heterogeneous entity and relational features are then separately fused to learn unified graph representations. Finally, we introduce RsConvE, a relation-specific extension of ConvE that restricts the prediction space during link inference, to estimate student-exercise recommendation scores and generate personalized exercises.

The main contributions are as follows.

- **Bridging the gap between knowledge tracing and exercise recommendation.** Existing learning-based recommendation methods estimate student knowledge states while largely overlooking exercise characteristics, creating a mismatch between learner representations and recommendation targets. We address this issue by incorporating exercise semantic and cognitive features into both student state modeling and recommendation, enabling recommendations that are better aligned with students' evolving knowledge states and learning needs.
- **A 2CKG4ER method based on content-aware cognitive knowledge graph.** We designed a feature extraction module, upon which the cognitive knowledge graph is built. A multi-source feature representation module then fuses features, followed by a knowledge graph embedding technique RsConvE to obtain recommendation scores. The recommendation reasons can also be provided based on the entity and relation features.
- **Empirical validation and interpretability analysis.** We compare 2CKG4ER with eight baseline recommendation methods on three educational datasets. Experimental results demonstrate that the proposed method achieves competitive adaptivity, with an approximately 2.9% improvement over the second-best baseline, while maintaining stable novelty performance. Moreover, 2CKG4ER provides interpretable recommendation reasons to enhance the transparency of personalized exercise recommendations.

2. Related Work

This section summarizes related work in three areas: personalized exerciser recommendation, knowledge graph embedding, and interpretable recommendation.

2.1. Personalized Exercise Recommendations

Exercise recommendation plays a crucial role in improving students' learning efficiency in online learning systems. Existing methods can be broadly classified into two categories: collaborative filtering-based methods and learning-based methods.

Collaborative filtering-based (CF) methods recommend exercises based on similarities among students or exercises. Walker et al. (2004) utilized learner response similarities for recommendation, while Thai-Nghe et al. (2010) applied matrix factorization to predict learner performance using user-item-rating triplets. Subsequent studies have incorporated additional student information, such as knowledge levels, learning styles, trustworthiness, and social ranking information, to enhance recommendation accuracy (Dwivedi and Bharadwaj, 2011; Segal et al., 2014; Wu et al., 2015; Guo et al., 2019). Although CF-based methods are simple and effective, they mainly rely on behavioral similarities from interaction data, failing to capture the complex, multi-relational structures among educational entities (e.g., students, exercises, and knowledge concepts). Furthermore, these methods typically overlook pedagogical factors and semantic contexts inherent to the learning process, including the semantic textual features of exercises, students' cognitive states and forgetting behaviors, and exercise difficulty. Consequently, they struggle to generate deeply personalized recommendations or offer pedagogical explanations for their decisions (see a summary in Table 1).

To improve educational personalization, learning-based methods use knowledge tracing (KT) models to estimate learners' knowledge states for generating tailored recommendations. For instance, Minn et al. (2018) combined deep KT models with K-means to dynamically cluster learners. Recently, reinforcement learning (RL)-based methods have emerged as a prominent trend in exercise recommendation, using leverage predicted knowledge states as reward signals to optimize exercise sequencing (Cheng et al., 2026; Zhang et al., 2024; Liu et al., 2019; Huang et al., 2019). For example, Liu et al. (2019) utilized the Deep Knowledge Tracing model (Piech et al., 2015) to simulate the environment by modeling student cognitive dynamics, subsequently applying RL to optimize recommendation strategies. Similarly, Cheng et al. (2026) adopted an actor-critic algorithm for exercise recommendation, using large language models to construct prerequisite and similarity graphs among concepts. Despite their performance, existing approaches typically rely on ID-based embeddings for graph entities and relations, neglecting valuable semantic content. To alleviate this issue, (Ozyurt et al., 2025) integrated a semantically grounded KT model to enrich question and knowledge concept representations with semantic information under a reinforcement learning framework. However, despite these advances, RL-based models suffer from black-box decision-making, lacking transparency regarding their underlying reasoning (see a summary in Table 1). To bridge this gap, this paper introduces an interpretable exercise recommendation method powered by a multi-relational cognitive knowledge graph.

2.2. Knowledge Graph Embedding

A knowledge graph is a multi-relational graph in which nodes represent entities and edges represent relations among entities. It is typically expressed as triples in the form $\langle \text{head entity}, \text{relation}, \text{tail entity} \rangle$. Due to their capability of representing structured factual information, knowledge graphs have been widely applied in recommendation systems (Xian et al., 2019; Yang et al., 2022). However, their complex structures and vast scales present significant computational challenges. To facilitate efficient representation and downstream reasoning, knowledge graph embedding (KGE) techniques are widely employed. KGE maps entities and relations into low-dimensional vector spaces while preserving their semantic properties and structural topologies (see (Cao et al., 2024) for a recent survey).

Various KGE techniques have been introduced using different scoring functions for estimating the triple plausibility scores. From this perspective, existing techniques can be classified into three groups: distance-based models, semantic matching models, and neural network models. 1) Distance-based models, a representative approach, define scoring functions based on geometric distances in the vector space. For instance, TransE (Bordes et al., 2013) models relations as translations between entity embeddings, while TransH (Wang et al., 2014) projects entities onto relation-specific hyperplanes to handle complex relational patterns. RotatE (Sun et al., 2019) further extends this idea by modeling relations as rotations in complex space, enabling effective handling of symmetry and inversion. Building on this, ReflectE (Zhang et al., 2022) in recent studies introduces reflection transformations over relation-specific hyperplanes to better capture the same relational patterns. Dense (Lu et al., 2022) models relations using non-commutative rotations and scaling in $SO(3)$ space to capture compositional and hierarchical semantics. CompoundE3D (Ge et al., 2023) unifies translation, rotation, and scaling operations into compositional geometric transformations for more expressive relation modeling. 2) Semantic matching models design scoring functions based on similarity or bilinear interactions in latent vector spaces. Representative methods include RESCAL (Nickel et al., 2011) and DistMult (Yang et al., 2015), which model relation-specific similarity through bilinear factorization, and ComplEx (Trouillon et al., 2016)

Table 1
Comparison of representative exercise recommendation methods.

Category	Representative Studies	Strengths	Limitations
CF-based methods	(Guo et al., 2019; Wu et al., 2015; Segal et al., 2014)	Simple to implement, and effective with abundant interaction data	Rely on shallow behavioral similarities, ignore students' cognitive states and exercise features
learning-based methods	(Ozyurt et al., 2025; Cheng et al., 2026; Zhang et al., 2024; Liu et al., 2019)	Model structural relationships of knowledge concepts, enabling adaptive recommendations	Limited interpretability of recommendation decisions
KGE-based methods	(Guan et al., 2023, 2025)	Leverage entity relations and provide recommendation rationales	Mainly ID-based embeddings and ignore features of exercises and students in learner modeling and recommendation

which maps embeddings to the complex domain to capture asymmetric relations. 3) Neural network models design scoring functions using nonlinear neural layers over entity and relation embeddings. ConvE (Dettmers et al., 2018) is a representative method that reshapes embeddings into 2D matrices and applies convolutional neural networks to capture local interaction patterns before computing triple scores. This enables flexible nonlinear interaction modeling beyond predefined geometric or bilinear assumptions.

KGE has shown promising results for exercise recommendation (Guan et al., 2023, 2025). Specifically, prior works (Guan et al., 2023, 2025) explored distance-based scoring functions to learn graph embeddings, demonstrating notable performance. However, distance-based scoring functions possess limited expressiveness compared to neural network architectures, which leverage non-linear layers for deep feature extraction. To address this limitation, this paper integrates RsConvE embedding model (a variant ConvE with tail-restricted prediction) to learn deep, non-linear interactions among entities and relations. Experimental results demonstrate that our approach outperforms these representative baseline models (Guan et al., 2023, 2025).

2.3. Interpretable Recommendation

Interpretable recommendation has drawn considerable attention in recent years (see (Zhang and Chen, 2020) for a survey). By offering the rationale behind algorithmic decisions, it enhances user satisfaction and trust in recommendation systems (Ma et al., 2024). Early interpretable approaches mainly focused on incorporating user features or item side-information to generate explanations, employing techniques such as attention mechanisms and recurrent neural networks (RNNs) (Dong et al., 2017). For example, Dong et al. (2017) generated product reviews from attribute-level information using neural models to produce coherent explanations. In recent studies, knowledge graphs have been increasingly utilized as they are helpful for generating intuitive explanations for recommended items (Xian et al., 2019; Yang et al., 2022). Because KGs contain rich information about users and items, they can effectively model complex relationships. Specifically, Xian et al. (2019) used multi-hop reasoning paths in knowledge graphs to justify and explicitly explain recommendation results. Yang et al. (2022) introduced contrastive learning for robust entity and relation representation in knowledge graph-based recommendation.

Although these approaches are effective in the e-commerce domain, they face limitations when applied to education for two reasons: (1) they fail to extract educational features from the learning process, and (2) multiple types of entities are involved, and the relationships among them are more complex than in the e-commerce domain. To address these issues, Guan et al. (2023, 2025) incorporated students' learning features, such as knowledge mastery levels, the likelihood of the subsequent concept, and forgetting behaviors, into knowledge graph-based recommendation. They created the knowledge graph with three entity types: knowledge concepts, students, and exercises. While this method provided interpretable exercise suggestions, it relied on ID-based representations of entities and relations, limiting its ability to capture rich semantic content (e.g., the textual content of exercises) and fine-grained pedagogical factors, such as learners' overall proficiency, exercise discrimination, and difficulty variations (see a summary in Table 1). To this end, we extract semantic content from exercises and capture pedagogical factors for both students and exercises.

Table 2

Summary of the notations.

Notation	Description
S, E, K	S : the set of learners, E : the set of exercises, and K : the set of knowledge concepts.
$\mathcal{L}_i$	$\mathcal{L}_i = \{(e_t, \{k_t\}, r_t) \mid t = 1, 2, \dots, T_i\}$: Learner s_i 's historical response sequence with exercise e_t , knowledge concept k_t , and the answer $r_t \in \{0, 1\}$.
$\mathcal{G} = (\mathcal{V}, \mathcal{R}, \mathcal{T}, \mathcal{X})$	A cognitive knowledge graph, where $\mathcal{V}$, $\mathcal{R}$, $\mathcal{T}$, and $\mathcal{X}$ denote the sets of entities, relation types, triples, and features, respectively.
$\mathcal{R} = R^m, R^f, R^r$	Mastery relation, forgetting relation, and recommendation relation in $\mathcal{G}$.
(h, r, t)	A triple of $\mathcal{T}$ with head entity, relation, and tail entity.
$\mathbf{x}_e^{\text{text}}, \mathbf{x}_e^{\text{cog}}$	Semantic and cognitive features of e , respectively.
θ_i	The ability parameter of student s_i .
$\text{cmkc}_{i,k}^{t+1}$	Predicted mastery level of the knowledge concept k .
$\text{cfr}_{i,e}^{t+1}$	The forgetting rate of exercise e at time $t + 1$.
δ_m, δ_f	Predefined mastery level and forgetting level, respectively.
W_{ij}	The recommendation relation score for building $\mathcal{G}$.
$\mathbf{X}_e, \mathbf{X}_r$	The concatenated features of exercises and cognitive relationships, respectively.
$\mathbf{f}_e, \mathbf{f}_s, \mathbf{f}_r$	The fused representation of exercise e , student s , and cognitive relationships $\{R^m, R^f\}$, respectively.
$\text{score}(s, \tilde{E})$	The recommendation scores over candidate exercises $\tilde{E}$ for student s .

We further learn fused embeddings that integrate semantic content and pedagogical factors into entity and relation representations, enabling more personalized exercise recommendations and interpretable recommendation reasons.

3. Problem formulation

This section defines the personalized learning recommendation task based on the cognitive knowledge graph. For clarity, Table 2 summarizes the main notations used in this paper.

3.1. Definitions

Definition 1. Given an online learning system with a set of students S , exercises E , and knowledge concepts K , the historical interaction sequence for a student s_i can be represented as $\mathcal{L}_i = \{(e_t, \{k_t\}, r_t) \mid t = 1, 2, \dots, T_i\}$, where $e_t \in E$ denotes the practised exercise at time t , $k_t \in K$ denotes e_t 's associated knowledge concept, $r_t \in \{0, 1\}$ indicates the response score, and T_i represents the total number of responses.

Definition 2. A cognitive knowledge graph is formally defined as an attributed heterogeneous directed multi-relational graph $\mathcal{G} = (\mathcal{V}, \mathcal{R}, \mathcal{T}, \mathcal{X})$, where $\mathcal{V} = S \cup E \cup K$ is a set of entities (i.e., nodes), $\mathcal{R} = \{R^m, R^f, R^r\}$ is a set of relation types, $\mathcal{T}$ contains relational facts in the form of edges, and $\mathcal{X} = \{\mathcal{X}_v, \mathcal{X}_e\}$ contains node attributes and edge attributes, respectively. Each edge is represented as a triple (h, r, t) , consisting of a head entity $h \in \mathcal{V}$, a relation $r \in \mathcal{R}$, and a tail entity $t \in \mathcal{V}$.

Regarding node attributes $\mathcal{X}$, exercise entities are associated with semantic features extracted from texts, while both student and exercise entities are associated with cognitive parameters derived from interaction sequences. For edge attributes, cognitive relations R^m and R^f are associated with scalar values, whereas R^r is unweighted. These three relation types are defined as follows:

- Cognitive mastery R^m : a relation between knowledge concepts and students, where $\text{cmkc}_{i,k} \in [0, 1]$ denotes the mastery level of student s_i on knowledge concept k .
- Cognitive forgetting R^f : a relation between exercises and students, where $\text{cfr}_{i,e} \in [0, 1]$ denotes the forgetting degree of student s_i on exercise e ;
- Recommendation R^r : a relation that exercises are recommended for students, which is computed according to students' cognitive states, including R^m and R^f .

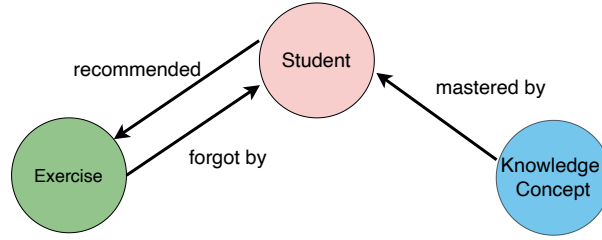


Figure 1: Cognitive Knowledge Graph

Consequently, the cognitive knowledge graph $\mathcal{G}$ comprises the following three types of triples: concept-student mastery triples $\{(k_j, r_{i,j}^m, s_i)\}$, exercises forgetting triples $\{(e_j, r_{i,j}^f, s_i)\}$, and exercises recommendation triples $\{(e_j, r_{i,j}^r, s_i)\}$. Figure 1 illustrates the structure of the cognitive knowledge graph, where nodes represent entities (students, knowledge concepts, and exercises), and edges represent three types of relationships.

3.2. Problem statement

Definition 3. Personalized learning recommendation aims to suggest exercises that align with students' cognitive needs, as inferred from their historical learning records, to enhance learning efficiency. Formally, given a recommendation size N , the task is to select a subset $E_S \subseteq E$ containing N exercises that the student has not yet practiced.

In the context of the cognitive knowledge graph, the recommendation task is formulated as a missing edge prediction problem. Given the head entity s (a student) and the relation R' (recommendation), the goal is to predict a tail entity (exercise e) that best matches the student's current cognitive states. To this end, a scoring function is introduced to evaluate the plausibility of each triple (s, R', e) , indicating the matching degree between the student and the exercise. These scores are then used to rank candidate exercises and generate the final recommendation list.

4. Method

This section presents the proposed 2CKG4ER for interpretable personalized exercise recommendation based on the content-aware cognitive knowledge graph. As shown in Figure 2, 2CKG4ER consists of four core modules: feature extraction, cognitive knowledge graph construction, feature representation, and recommendation. (1) Initially, the feature extraction module extracts features from the learning process, including entity features (the semantic content of exercises and cognitive parameters) and relational features (mastery and forgetting). (2) Building on these extracted features, the construction module extract recommendation triples. The constructed content-aware cognitive knowledge graph $\mathcal{G}$ composed of three entity types (students, exercises, and knowledge concepts), and three relationship types, including mastery, forgetting and recommendation. (3) The representation module learns the representation of entities and relations by using an MLP operator to fuse multi-modal features. (4) The recommendation module calculates the recommendation scores between students and exercises via ReConvE, a variant of the ConvE knowledge graph embedding model, and infers the reasons behind the recommendation decisions from the knowledge graph $\mathcal{G}$.

4.1. Feature Extraction

4.1.1. Feature Extraction for Entities

This component extracts semantic content of exercises alongside the cognitive parameters of both exercises and students. Many exercise recommendation methods treat exercises merely as indices, overlooking these crucial entity features. This neglect makes recommendation less practical, as exercises associated with the same knowledge concept can vary significantly in textual semantics, difficulty levels, and discrimination. In contrast, our proposed 2CKG4ER method encodes these entity features to ensure recommendation practical and personalized.

(1) Semantic Content Extraction of Exercises: we adopt bidirectional gated recurrent unit (BiGRU) networks to capture text-based semantic features of exercises and. This process consists of two steps: preprocessing and text sequence representation. Preprocessing involves segmenting words and removing stopping words. Let text_e represent the raw exercise text, the preprocessed of exercise texts is denoted as text_e . Text sequence representation uses a BiGRU

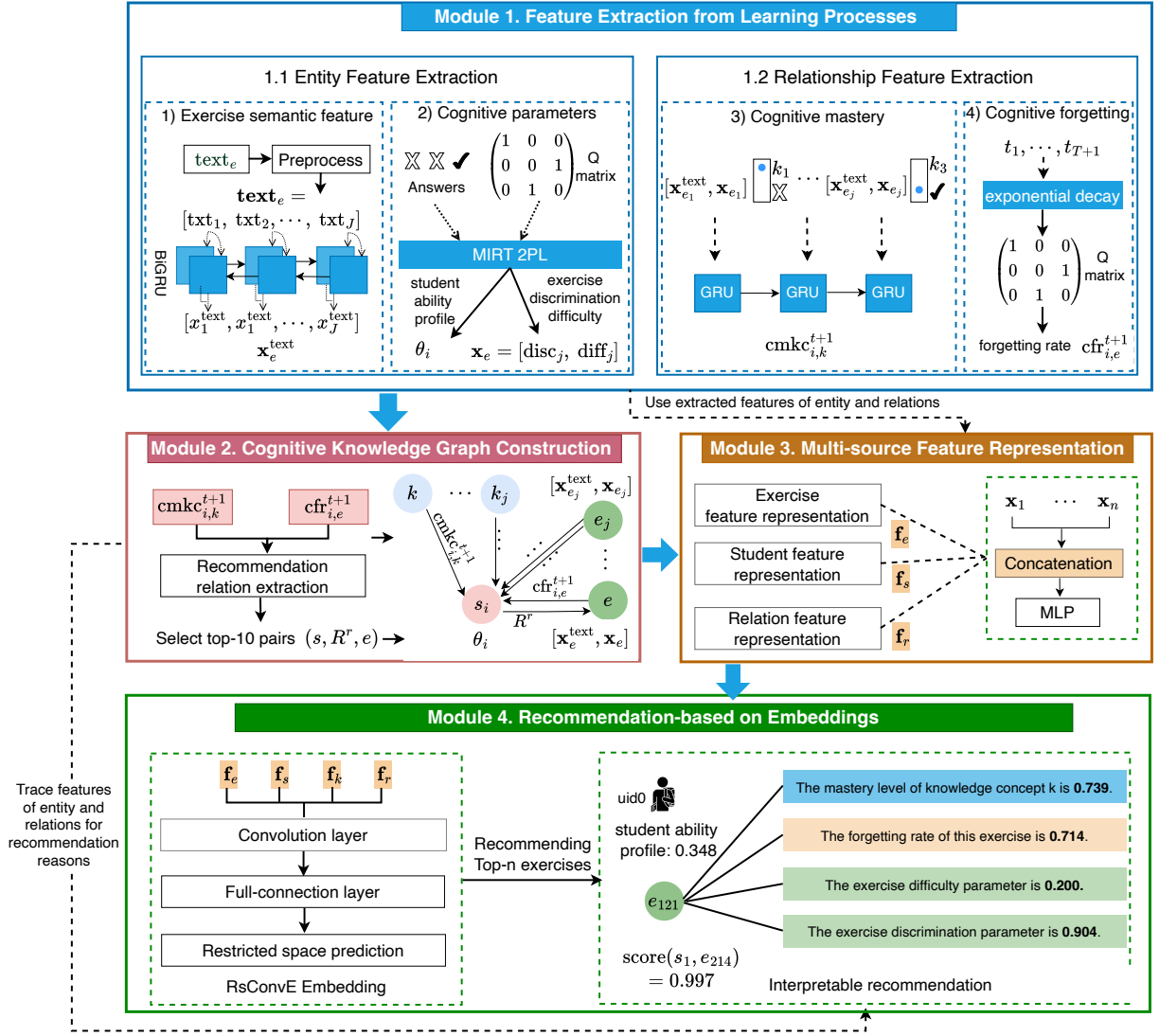


Figure 2: The overall framework of 2CKG4ER.

to learn the embedding of text_e by combining hidden states from forward and backward GRUs. The resulting exercise text embedding is formulated as follows:

$$\mathbf{x}_e^{\text{text}} = [\bar{\mathbf{h}}_{e,t}^{\text{text}}, \bar{\mathbf{h}}_{e,t}^{\text{text}}], \text{ where } \bar{\mathbf{h}}_{e,t}^{\text{text}} = \text{GRU}_f(\text{text}_e, \bar{\mathbf{h}}_{t-1}), \bar{\mathbf{h}}_{e,t}^{\text{text}} = \text{GRU}_b(\text{text}_e, \bar{\mathbf{h}}_{t+1}). \quad (1)$$

2) Cognitive Parameter Extraction for Exercises and Students: To extract cognitive features, we calibrate interaction sequences using the two-parameter logistic (2PL) Item Response Theory (IRT) model. We employ the MIRT (Chalmers, 2012) to align students' latent ability dimensions with knowledge concepts. Under this way, the probability that student s_i responds correctly to exercise e_j is modeled as:

$$P(X_{ij} = 1 | \theta_i, \text{disc}_j, \text{diff}_j) = \frac{1}{1 + \exp(-1.7 \text{disc}_j(\theta_i - \text{diff}_j))}, \quad (2)$$

$$\mathbf{x}_{e_j}^{\text{cog}} = \text{concat}(\text{disc}_j, \text{diff}_j),$$

where θ_i denotes the overall ability profile of student s_i , representing s_i 's overall proficiency level; disc_j denotes the discrimination parameter of exercise e_j , representing the exercise effectiveness in differentiating between students of high versus and low ability; and diff_j denotes the exercise difficulty parameter, with high values indicating cognitively demanding. $\mathbf{x}_e^{\text{cog}}$, representing the cognitive parametric representation of exercise e_i , is obtained by concatenating discrimination parameter disc_j and difficulty parameter diff_j .

4.1.2. Feature Extraction for Relationships

This component extracts two types of cognitive relation features, cognitive mastery and cognitive forgetting. (1) Cognitive mastery reflects students' evolving knowledge states, which are obtained via a knowledge tracing (KT) model from their historical interaction sequences. Inspired by MMKT (Wang et al., 2025b), we develop an Exercise Content-Aware KT (ECAKT) model that incorporates exercise textual semantics and cognitive parameters. We exclude the image modality and text-image fusion module because only a small proportion of exercises contain images. Unlike conventional KT models that solely rely on knowledge concept IDs, ECAKT preserves exercise-level semantic information. This design bridges the gap between concept-level cognitive state modeling in KT and exercise-level recommendation, enabling 2CKG4ER to better distinguish exercises associated with the same knowledge concept but varying in semantic content, difficulty, and discrimination, thereby improving personalized exercise recommendations. (2) Cognitive forgetting quantifies the rate at which a learner fails to recall previously attempted exercises. It is calculated using an exponential decay function based on the elapsed time intervals.

The proposed recommendation method is not limited to this specific knowledge tracing model. The cognitive knowledge graph builds on the generated features of nodes and edges independent of the internal implementation of the chosen knowledge tracing model (for mastery). Therefore, other advanced knowledge tracing models can also be integrated into our proposed method if they provide learner-state estimates of mastery and forgetting.

3) Cognitive Mastery Extraction: Cognitive mastery characterizes the student's ability to correctly answer exercises related to specific knowledge concepts. This feature is acquired by the ECAKT model. Given the historical interaction sequence $\mathcal{L}_i$, exercise semantic feature $\mathbf{x}_e^{\text{text}}$ and cognitive feature $\mathbf{x}_e^{\text{cog}}$, the student's cognitive state h_t is obtained using a GRU network as follows:

$$h_t = \text{GRU}(h_{t-1}, \mathbf{x}_{e,t}), \text{ where } \mathbf{x}_{e,t} = [\mathbf{x}_e^{\text{text}}, \mathbf{x}_e^{\text{cog}}, \text{Onehot}(k), r_t]. \quad (3)$$

The probability $\hat{r}_{t+1}$ of correctly answering the next exercise is predicted by

$$\hat{r}_{t+1} = \sigma(\mathbf{W}_r [h_t, \text{Onehot}(k_{t+1})] + \mathbf{b}_r), \quad (4)$$

where σ is the sigmoid function, $\mathbf{W}_r$ and $\mathbf{b}_r$ are trainable weight and bias parameters, respectively. We use $\hat{r}_{t+1}$ as the mastery level of knowledge concept k_{t+1} , namely $\text{cmkc}_{i,k}^{t+1} = \hat{r}_{t+1}$.

The training objective is to minimize the difference between the predicted mastery level $\text{cmkc}_{i,k}^{t+1}$ and students' actual performance r_{t+1} by using the binary cross-entropy loss:

$$\mathcal{L}_{\text{cme}} = - \sum_{i=1}^{|S|} \sum_{t=1}^{T_i-1} \left(r^{t+1} \log \text{cmkc}_{i,k}^{t+1} + (1 - r^{t+1}) \log (1 - \text{cmkc}_{i,k}^{t+1}) \right). \quad (5)$$

4) Cognitive Forgetting Extraction: The forgetting curve suggests that students tend to forget knowledge concepts over time. This decline highlights the importance of regular review to maintain knowledge retention. Recent studies have proven the effectiveness of incorporating forgetting mechanisms in knowledge tracing (Bai et al., 2025) and exercise recommendation (Yang et al., 2026). To capture the forgetting of knowledge concepts, we utilize the exponential decay function (Averell and Heathcote, 2011), which is widely adopted to fit the forgetting curve. The overall cognitive forgetting rate $\text{cfr}_{i,e}^{t+1}$ is obtained by aggregating the forgetting contribution of each knowledge concept that are associated with an exercise, which is calculated as follows:

$$\begin{aligned} \text{cfr}_{i,e}^{t+1} &= \frac{1}{|K|} \sum_{k \in K} \text{decay}_{i,e} \cdot \mathcal{Q}_{e,k}, \\ \text{s.t.}, \text{decay}_{i,e} &= 1 - e^{-\alpha \cdot \Delta t}, \end{aligned} \quad (6)$$

Table 3Entity feature of Cognitive Knowledge Graph $\mathcal{G}$.

Entity	Node Attribute	Description	Source
Student	Cognitive parameter θ_i	The ability profile of student s_i	Interaction sequences
Exercise	Semantic vector $\mathbf{x}_e^{\text{text}}$	The semantic content of exercise e	Exercise text
Exercise	Cognitive parameter $\mathbf{x}_e^{\text{cog}}$	The difficulty and discriminative parameters	Interaction Sequence

where $Q_{e,k}$ is the exercise-knowledge concept association matrix, indicating whether exercise e assesses knowledge concept k , and K is the subset of knowledge concepts. Notation Δt denotes the time elapsed since student s_i last interacted with exercise e , and α is a forgetting parameter that controls the rate of memory decay.

4.2. Cognitive Knowledge Graph Construction

The cognitive knowledge graph $\mathcal{G}$ for the 2CKG4ER contains three types of entities (i.e., students, exercises, and knowledge concepts) and three types of relationships (cognitive mastery, cognitive forgetting, and recommendation). For the entity, as shown in Table 3, we extract semantic content $\mathbf{x}_{e,t}^{\text{text}}$ from the exercise text and cognitive parameters (i.e., $\mathbf{x}_e^{\text{cog}}$ and θ_i for student s_i). Among the relationships, the cognitive mastery $\text{cmkc}_{i,k}^{t+1}$ is extracted from student interaction sequences according to GRU networks. The cognitive forgetting $\text{cfr}_{i,e}^{t+1}$ is derived from timestamps of the interaction sequences by an exponential decay function.

6) Recommendation Relation Extraction: The final relationship we define is the recommendation link between students and exercises, representing whether exercises match a student's current learning progress. The distance score $W_{i,j}$ between student s_i and excises e_j is computed based on the two relational features ($\text{cmkc}_{i,k}^{t+1}$ and $\text{cdr}_{k,t+1}$) as follows:

$$W_{i,j} = \sqrt{\left(\delta_m - \prod_{k \in K_e} \text{cmkc}_{i,k}^{t+1}\right)^2 + \left(\delta_f - \text{cfr}_{i,e}^{t+1}\right)^2}, \quad (7)$$

where the first component $(\delta_m - \prod_{k \in K_e} \text{cmkc}_{i,k}^{t+1})$ represents mastery match distance, where δ_m is the predefined target mastery level required for recommending an exercise. The second component $(\delta_f - \text{cfr}_{i,e}^{t+1})^2$ represents the forgetting matching distance where δ_f is the predefined target forgetting level. Smaller values across both components result in a lower distance score $W_{i,j}$.

After calculating $W_{i,j}$ for all student-exercise pairs, we rank exercises in ascending order. Because the candidate pool of exercises is extremely large, we identify top-n pairs of students and exercises with the lowest $W_{i,j}$ to build the recommendation triples. Specifically, in our experiments, to follow (Guan et al., 2023, 2025)'s work, the top-10 pairs of students and exercises are selected for each student to construct positive recommendation triples as follows:

$$R_{\text{positive}}^r = \{(s_i, R^r, e_j) \mid W_{ij} \in \text{Top}_{10}(\{W_{ij}\}), s_i \in S, e_j \in E\} \quad (8)$$

where Top_{10} returns the 10 smallest weights. In these triples, students and exercises act as nodes in the cognitive knowledge graph, connected by recommendation edges without attributes. Negative recommendation samples are generated by randomly replacing the tail exercise entities in positive recommendation triples with unobserved exercises.

Table 4 shows triples and edge attributes of the cognitive knowledge graph. Please note that only the cognitive mastery and forgetting relationships are associated with edge weights. For the recommendation relationship, we do not maintain $W_{i,j}$ as the recommendation strength because it is not directly extracted from the interaction sequences. Instead, we use $W_{i,j}$ to construct recommendation triples.

4.3. Multi-source Feature Representation

Conventional knowledge graph embedding methods represent entities and relations solely by trainable ID embeddings, which ignore the rich semantic and cognitive information available in educational recommendation. To address this limitation, we learn feature representations for entities and relations from multiple information sources. These feature representations serve as inputs for the next section.

Table 4
Triples and Edge Attributes of Cognitive Knowledge Graph $\mathcal{G}$.

Head	Relation Type	Tail	Edge At-tribute	Description
Knowledge concept	Cognitive mastery R^m	Student	$\text{cmkc}_{i,k}^{t+1}$	Student s_i 's mastery level on the knowledge concept
Exercise	Cognitive forgetting R^f	Student	$\text{cfr}_{i,e}^{t+1}$	Student s_i 's forgetting degree on the exercise
Student	Recommendation R^r	Exercise	-	The exercise is recommended to the student

4.3.1. Multi-source Entity Representation Fusion

To incorporate heterogeneous information into knowledge graph representation learning, we construct feature representations for students, exercises, and relations prior to graph embedding. Specifically, all heterogeneous features are first concatenated, followed by feature fusion using an MLP operator.

Exercise Feature Representation: As shown in Table 3, each exercise is characterized by textual semantics $\mathbf{x}_e^{\text{text}}$ and cognitive parameters $\mathbf{x}_e^{\text{cog}}$. Since these features capture complementary information, they are concatenated and fused via an MLP operator, calculated as follows:

$$\begin{aligned} \mathbf{X}_e &= [\mathbf{x}_e^{\text{text}}, \mathbf{x}_e^{\text{cog}}] \in \mathbb{R}^{2d_1} \\ \mathbf{f}_e &= \text{MLP}(\mathbf{X}_e) \in \mathbb{R}^{d_1}, \end{aligned} \quad (9)$$

where $\mathbf{X}_e$ denotes the concatenated feature. After concatenation, an MLP is applied to learn the interactions between semantic, cognitive, and ID features for a unified feature representation $\mathbf{f}_e$.

Student Feature Representation: Similar to the exercise representation, the student feature vector $\mathbf{f}_s \in \mathbb{R}^{d_1}$ is obtained by projecting the student's ability parameter θ_i into the common latent space using MLP operations:

$$\mathbf{f}_s = \text{MLP}(\mathbf{x}_i^\theta) \in \mathbb{R}^{d_1}, \quad (10)$$

where $\mathbf{x}_i^\theta$ is the student's cognitive ability.

4.3.2. Multi-source Relation Representation Fusion

As shown in Table 4, each cognitive relation consists of two components: a semantic relation description (see mastery definition and forgetting definition) and a continuous cognitive weight. The semantic meaning is represented by a BiGRU while the weight undergoes a linear transformation. We then concatenate the representation of type and weight as follows:

$$\begin{aligned} \mathbf{x}_r^{\text{type}} &= \text{BiGRU}(\text{discrip}_r), \text{ s.t., } r \in \{\text{mastery, forgetting}\} \\ \tilde{\mathbf{x}}_r^{\text{str}} &= \mathbf{W}_r c_r + \mathbf{b}_r, \text{ s.t., } c_r \in \{\text{cmkc}_{i,k}^{t+1}, \text{cfr}_{i,e}^{t+1}\} \\ \mathbf{X}_r &= [\mathbf{x}_r^{\text{type}}, \tilde{\mathbf{x}}_r^{\text{str}}] \in \mathbb{R}^{2d_1}. \end{aligned} \quad (11)$$

where discrip_r denotes the text description of relations following Definition 2, and $\mathbf{W}_r \in \mathbb{R}^{d_1 \times 1}$ and $\mathbf{b}_r \in \mathbb{R}^{d_1}$ are trainable parameters. An MLP operator is subsequently applied to the combined relation feature vector $\mathbf{x}_r$ to obtain the unified relation representation:

$$\mathbf{f}_r = \text{MLP}(\mathbf{X}_r) \in \mathbb{R}^{d_1}. \quad (12)$$

For the recommendation relation, which contains only semantic meaning without an associated weight, the relation feature representation is directly obtained from the BiGRU and then projected via an MLP operator.

4.4. Recommendation based on embeddings

This module applies the knowledge graph embedding techniques to learn the low-dimensional embeddings of entities and relations, which are subsequently used to recommend exercises. Since the ConvE model (Dettmers et al., 2018) is expressive and can learn deep features among entities and relations, we adapt it to represent our graph. Because the valid tail entities depend strictly on the relation type, we restrict the tail entity space to the specific target entity type during prediction, introducing a variant we denote as Relation-specific ConvE (RsConvE). Finally, the recommendation scores between students and exercises are inferred using the trained RsConvE model.

4.4.1. Restricted-space ConvE Learning

Given a graph triple (h, r, t) , the head and relation embeddings are shaped into two-dimensional feature maps, followed by a convolution layer to model their interactions:

$$\mathbf{v} = f \left(\text{vec} \left(\sigma \left(\left[\bar{\mathbf{f}}_h, \bar{\mathbf{f}}_r \right] * \mathbf{\Omega} \right) \right) \mathbf{W} \right), \quad (13)$$

where $\bar{\mathbf{f}}_h$ and $\bar{\mathbf{f}}_r$ denote the reshaped head embedding and relation embeddings, respectively; $\mathbf{\Omega}$ is the convolution kernel; vec denotes the vectorization operator; and $f(\cdot)$ represents the nonlinear activation function.

The probability distribution over the candidate tail entities is computed as follows:

$$\mathbf{p} = \sigma \left(\mathbf{v} \mathbf{Z}_t^\top \right), \quad (14)$$

where $\mathbf{Z}_t$ denotes the embedding matrix of all candidate tail entities, $\sigma(\cdot)$ is the sigmoid activation function, and $\mathbf{p}$ represents the matching scores between head entities and tail entities.

Restricted-Space Tail Prediction: As shown in Table 4, the candidate tail entities depend strictly on the relation type. For cognitive relations, namely mastery, and forgetting, the candidate tails are student entities, whereas for the recommendation relation, the candidate tails are exercise entities. Specifically, cognitive relations are optimized using the conventional ConvE multi-label learning objective (Dettmers et al., 2018). Accordingly, for each cognitive triple (h, r, s) with $r \in \{R^m, R^f\}$, the model predicts all candidate student entities simultaneously, where the ground-truth labels are represented as a multi-hot vector. The cognitive relation loss is defined as

$$\mathcal{L}_{\text{cog}} = - \sum_{(h,r,s) \in \mathcal{G}_{\text{cog}}} \frac{1}{|S|} \sum_s \left[y_s \log p_s + (1 - y_s) \log(1 - p_s) \right], \text{ s.t., } h \in K \cup E, s \in S \quad (15)$$

where p_s denotes the predicted probability of student entity s , and $y_s \in \{0, 1\}$ is the corresponding element of the multi-hot target vector, where $y_s = 1$ indicates that the triple (h, r, s) exists and $y_s = 0$ indicates that the triple (h, r, s) does not exist.

For the recommendation relation (s, R^r, e) , since the extracted positive triples only contains partial exercises, we adopt negative sampling during training. For each positive recommendation triple, five unobserved exercises are randomly sampled as negative instances, yielding the recommendation loss

$$\mathcal{L}_{R^r} = - \sum_{(s,R^r,e) \in \mathcal{G}_{R^r} \cup \mathcal{G}_{R^r}^-} \left[y \log p + (1 - y) \log(1 - p) \right], \quad (16)$$

where p denotes the predicted probability of the sampled exercise, $\mathcal{G}_{R^r}^-$ contains the negative sampled triples, and $y \in \{0, 1\}$ indicates whether the recommendation triple is observed or sampled.

Finally, the two objectives are jointly optimized:

$$\mathcal{L} = \mathcal{L}_{\text{cog}} + \mathcal{L}_{R^r}. \quad (17)$$

4.4.2. Exercise Scoring

For each student, the features of entities and relationships are extracted (as described in Section 4.1), followed by the construction of triples associated with student node s (as described in Section 4.2). Then, the multi-source representation of these entity and relationship features is fused separately, generating the input for the RsConvE model. After training, the RsConvE model provides a scoring function (see Equation (14)) between head entities and tail entities.

Regarding the recommendation relationship, the recommendation scores between student s and candidate exercises $\tilde{E}$ are formulated as follows:

$$\text{score}(s, \tilde{E}) = \sigma \left(f \left(\text{vec} \left(\sigma \left(\left[\bar{\mathbf{f}}_s, \bar{\mathbf{f}}_{R^r} \right] * \Omega \right) \right) \mathbf{W} \right) \mathbf{Z}_{\tilde{E}}^T \right), \quad (18)$$

where $\mathbf{f}_s$ and $\bar{\mathbf{f}}_{R^r}$ denote the reshaped 2D feature maps of the student embedding and the recommendation relation embedding, respectively; $\mathbf{Z}_{\tilde{E}}$ denotes the embedding matrix of candidate tail entities $\tilde{E}$. This score reflects the match degree between an exercise $\tilde{e} \in \tilde{E}$ and student $s \in S$ from the perspective of the entire cognitive knowledge graph $\mathcal{G}$. We sort exercises in descending order recommendation scores and select the top- N exercises that the student has not previously attempted to form the final recommendation list.

After the recommendation list is generated, we provide the underlying reasons for each exercise recommendation based on the cognitive knowledge graph $\mathcal{G}$. The exercise difficulty parameter diff_e , discrimination parameter disc_e , and student ability parameter θ_i , along with the two cognitive relational features ($\text{cmkc}_{i,k}^{t+1}$ and $\text{cfr}_{i,e}^{t+1}$) are used to explain the recommendation decisions. As illustrated in Fig. 2, exercise e_{121} is recommended to a student with an ability level of 0.348, while the exercise difficulty and discrimination parameters are 0.200 and 0.904, respectively. Furthermore, the student's mastery level of related knowledge concept k are 0.739, and the forgetting rate of this exercise is 0.714. These recommendation evidences indicate that the recommendation of an easy exercise is attributed to the learner's relatively low ability level. Similarly, the high discrimination value (0.904) of the recommended exercise is acceptable because the students has achieved a relatively high mastery level (0.814) of the corresponding knowledge concept. Consequently, while RsConvE is used to produce the ranking scores, the cognitive knowledge graph provides traceable educational evidence to interpret why the recommended exercise matches the student's current cognitive states.

5. Experiments

This section evaluates the proposed 2CKG4ER on five real-world educational datasets and addresses three research questions:

RQ1: Does the proposed 2CKG4ER achieve better performance compared with baselines??

RQ2: How do the extracted entity features and relational features affect the performance of the proposed 2CKG4ER?

RQ3: Are the exercise recommendation reasons of 2CKG4ER effective and rational?

5.1. Datasets

To evaluate the effectiveness of the proposed method across different educational recommendation scenarios, we conduct experiments on three public educational datasets: Eedi-sub, Algebra 2005, and XES3G5M-sub. These datasets contain questions from different subject domains and include both English- and Chinese-language data. All datasets contain student response records and exercise-knowledge concept associations, which are required for constructing the cognitive knowledge graph.

- Eedi-sub is derived from the Eedi2020 dataset released in the NeurIPS 2020 Education Challenge (Wang et al., 2020, 2021). The original Eedi2020 dataset contains 1,382,727 responses from 4,918 students on 948 multiple-choice mathematics exercises. For the exercise recommendation task, we retain learners whose response sequence length is between 10 and 200, resulting in the Eedi-sub dataset.
- **Algebra 2005** is from the KDD Cup 2010 Educational Data Mining Challenge (Stamper et al., 2010), which contains step-level tutoring records with correctness labels and knowledge-component annotations.
- **XES3G5M-sub** is derived from the a large-scale mathematics XES3G5M benchmark dataset (Liu et al., 2023), which has 18,066 learners, 7,652 questions and 865 knowledge concepts. To keep computationally feasible, we construct XES3G5M-sub by keeping sequences whose length is between 10 and 400.

Table 5 summarizes the statistics of the datasets used in the experiments. Here, “interactions” denotes student response records, while “triples” denotes graph edges constructed from mastery, forgetting, and recommendation relations.

Table 5
Statistics of datasets

Dataset	KCs	Students	Exercises	Interactions	Entities	Triples
Eedi-sub	57	935	948	98,742	1,940	946,685
Algebra 2005	112	574	1,084	607,025	1,770	690,804
XES3G5M-sub	385	968	1,606	331,119	2,959	1,934,558

5.2. Baselines

We compare the proposed 2CKG4ER method with eight baseline methods from two categories: five KGE-based methods (including TransE-ER, TransE-adv-ER, RotatE-ER, DistMult-ER, and ComplEx-ER) and three traditional CF-based recommendation methods (including EB-CF, SB-CF, and CBF). These five KGE-based methods use three relational features (mastery, the likelihood of the subsequent concepts, and forgetting) by ID-based embeddings as introduced in Guan et al. (2023, 2025), whereas CF-based methods use only interaction sequences. The KGE-based baselines provide a direct comparison with existing knowledge graph embedding approaches, while the CF-based baselines evaluate the effectiveness of incorporating knowledge graphs to exercise recommendation. All methods are evaluated using the same dataset partition.

- **TransE-ER** (Guan et al., 2025): TransE models relations as translation vectors from head entities to tail entities. In this baseline, TransE is trained on the cognitive knowledge graph by converting continuous relation features into discrete relation types and relying on ID-based embeddings.
- **TransE-adv-ER** (Guan et al., 2025): TransE-adv enhances TransE with self-adversarial negative sampling, assigning higher weights to hard negative triples with larger prediction scores to learn more discriminative embeddings.
- **RotatE-ER** (Sun et al., 2019): RotatE models relations as rotations in the complex vector space, examining the effectiveness of rotational relation modeling for cognitive graph-based exercise recommendation.
- **DistMult-ER** (Yang et al., 2015): DistMult is a bilinear semantic matching model that scores triples through a diagonal bilinear product. It provides a representative real-valued semantic matching baseline for graph-based exercise recommendation.
- **ComplEx-ER** (Trouillon et al., 2016): ComplEx extends bilinear modeling to the complex vector space and captures asymmetric relations. This baseline evaluates complex-valued relation representations for learner-exercise recommendation.
- **EB-CF** (Sarwar et al., 2001): Exercise-based collaborative filtering recommends exercises based on exercise similarity derived from learner response histories. It serves as a non-KG baseline based on behavioral similarity.
- **SB-CF** (Liu and Hao, 2012): Student-based collaborative filtering recommends exercises based on the behavior of similar learners. It represents a learner-neighborhood recommendation strategy without knowledge graph information.
- **CBF** (van Meteren and van Someren, 2000): Content-based filtering recommends exercises based on exercise-knowledge concept associations. It ranks candidate exercises by matching their knowledge concepts with the learner's historical knowledge profile.

5.3. Evaluation Metrics

Exercise recommendation is inherently personalized, and the datasets lack ground-truth labels for recommendation correctness. Therefore, we evaluate recommendation quality from an educational perspective using two metrics: Adaptability (Ada) and Novelty (Nov), following (Guan et al., 2023, 2025). The original Accuracy metric is renamed as Adaptability because it does not represent prediction accuracy in the absence of explicit recommendation labels.

Adaptability (Ada): Adaptability measures how well the difficulty of recommended exercises matches a student's current knowledge state. For the recommended exercise set E_s for student s_i , the adaptability is calculated as:

$$\text{Ada} = \frac{1}{|E_s|} \sum_{e \in E_s} \frac{1}{|K_e|} \sum_{k \in K_e} \left(1 - \left| \text{cmkc}_{i,k}^{t+1} - \delta_m \right| \right), \quad (19)$$

where $K_e = \{k \in K \mid Q(k, e) = 1\}$ represents the set of knowledge concepts associated with exercise e , and δ_m is the target mastery level.

Novelty (Nov): Novelty evaluates whether the recommended exercises contain new or infrequently practiced knowledge concepts. Given K_s^{hist} as the set of knowledge concepts that student s has previously answered correctly, and $K_s^{R^r}$ represent the set of concepts associated the recommended exercises E_s . To account for forgetting over time, we assign a time-dependent weight to each previously mastered knowledge concept as follows:

$$w_{s,k} = \exp(-\mu \Delta t_{s,k}), \quad (20)$$

where $\Delta t_{s,k}$ is the time elapsed since the last correct response of student s on knowledge concept k , and $\mu > 0$ a predefined temporal decay parameter. The novelty score is then calculated as:

$$\text{Nov} = 1 - \frac{\sum_{k \in K} \min \left(w_{s_i,k} \cdot \mathbb{I}(k \in K_{s_i}^{\text{hist}}), w_{s_i,k} \cdot \mathbb{I}(k \in K_{s_i}^{R^r}) \right)}{\sum_{k \in K} \max \left(w_{s_i,k} \cdot \mathbb{I}(k \in K_{s_i}^{\text{hist}}), w_{s_i,k} \cdot \mathbb{I}(k \in K_{s_i}^{R^r}) \right)}, \quad (21)$$

where $\mathbb{I}(\cdot)$ is the indicator function, and K is the set of all knowledge concepts. A higher Nov value indicates that the recommended exercises cover more unseen or long-unpracticed knowledge concepts.

5.4. Implementation Details

We split all datasets at a ratio of 75% and 25% for training and testing, respectively. In the feature extraction module, the target mastery level σ_m and the target forgetting level σ_f are both set to 0.8. We chose a higher value of σ_m to favor easier exercises. According to Papoušek et al. (2016), easy exercises yield better learning outcomes during shorter study sessions (10% error rate), whereas harder exercises (50% error rate) require longer engagement periods. Regarding σ_m , since short-term retention gradually builds long-term memory (Kitamura et al., 2017), we set this parameter to prioritize recommending exercises that students are most at risk of forgetting. For each student s_i in the training set, the top-10 exercises with the smallest distances $W_{i,j}$ (see Equation (8)) are selected to construct positive recommendation triples (s_i, R_r, e_j) . When training the 2CKG4ER, we generate five negative samples for each positive recommendation triple by randomly replacing the tail entity with an unpracticed exercise. The existing positive cognitive triples remain unchanged during this process.

The entity and relation embedding dimension is set to 200. Accordingly, the inputs to the convolutional layer in RsConvE are reshaped into 20×10 2D feature maps. The batch size is 1024, the learning rate is 0.001, and the model is trained using the Adam optimizer for 25 epochs. The dropout rates for the input layer, hidden layer, and feature maps are set to 0.2, 0.2, and 0.3, respectively. During testing, all recommendation triples are excluded from the evaluation graph to avoid data leakage. The recommendation scores for unobserved student-exercise pairs are computed solely the embeddings generated from the trained feature-fusion layers via Equation (18).

The embedding dimension of the other five KGE-based baselines is 1000, following Guan et al. (2025)'s work. All KGE-based methods are performed five times on each training dataset using five different random seeds (namely 2024, 2025, 2026, 2027, and 2028), and we report both the mean and standard deviation of the results. The CF-based methods are run once under the same data partition since they are deterministic. We set the recommendation length to $N = \{5, 10, 15, 20, 25, 30, 35, 40, 45, 50\}$. All experiments are implemented in Python using PyTorch and NumPy, and the source code of our 2CKG4ER is available at <https://github.com/Tangfgjk/KG4ER>.

5.5. Results

5.5.1. Overall Performance Comparison (RQ1)

Experiment 1 compares our proposed 2CKG4ER with eight baselines regarding two metrics: Adaptability and Novelty. Tables 6, 7, and 8 report the comparison results across three datasets under varying recommendation sizes $N = \{5, 10, 15, 20, 25, 30, 35, 40, 45, 50\}$. Experimental results reveal four key insights.

Table 6

Performance comparison of different methods on Eedi-sub under varying recommendation sizes N (Mean $\pm$ Std).

(a) Adaptability performance (Ada).

N	EB-CF	SB-CF	CBF	RotatE	DistMult	ComplEx	TransE	TransE-adv	2CKG4ER
5	0.690	0.690	0.576	0.695 ± 0.002	0.693 ± 0.003	0.697 ± 0.003	0.705 ± 0.006	0.703 ± 0.004	0.722 ± 0.006
10	0.687	0.690	0.577	0.694 ± 0.001	0.693 ± 0.003	0.696 ± 0.002	0.704 ± 0.005	0.703 ± 0.004	0.723 ± 0.001
15	0.689	0.691	0.579	0.694 ± 0.000	0.692 ± 0.002	0.695 ± 0.003	0.705 ± 0.003	0.703 ± 0.003	0.723 ± 0.003
20	0.689	0.690	0.581	0.693 ± 0.000	0.691 ± 0.002	0.695 ± 0.003	0.705 ± 0.003	0.702 ± 0.002	0.722 ± 0.004
25	0.689	0.690	0.584	0.693 ± 0.001	0.691 ± 0.002	0.695 ± 0.002	0.705 ± 0.002	0.702 ± 0.002	0.722 ± 0.003
30	0.689	0.690	0.586	0.693 ± 0.001	0.691 ± 0.002	0.695 ± 0.002	0.705 ± 0.001	0.702 ± 0.001	0.721 ± 0.002
35	0.689	0.689	0.588	0.693 ± 0.001	0.692 ± 0.002	0.695 ± 0.002	0.704 ± 0.001	0.701 ± 0.002	0.720 ± 0.001
40	0.689	0.689	0.590	0.693 ± 0.001	0.692 ± 0.002	0.694 ± 0.002	0.704 ± 0.001	0.702 ± 0.002	0.720 ± 0.002
45	0.689	0.689	0.592	0.693 ± 0.001	0.692 ± 0.002	0.694 ± 0.001	0.703 ± 0.001	0.702 ± 0.002	0.719 ± 0.001
50	0.689	0.689	0.594	0.693 ± 0.001	0.692 ± 0.002	0.694 ± 0.001	0.703 ± 0.002	0.701 ± 0.002	0.718 ± 0.001

(b) Novelty performance (NOV).

N	EB-CF	SB-CF	CBF	RotatE	DistMult	ComplEx	TransE	TransE-adv	2CKG4ER
5	0.981	0.980	0.997	0.997 ± 0.000	0.995 ± 0.003	0.998 ± 0.001	0.996 ± 0.002	1.000 ± 0.000	0.982 ± 0.007
10	0.985	0.982	0.997	0.997 ± 0.000	0.994 ± 0.003	0.998 ± 0.001	0.995 ± 0.002	1.000 ± 0.000	0.982 ± 0.003
15	0.985	0.982	0.997	0.998 ± 0.000	0.994 ± 0.003	0.998 ± 0.000	0.995 ± 0.002	0.999 ± 0.000	0.981 ± 0.002
20	0.984	0.982	0.996	0.997 ± 0.000	0.994 ± 0.003	0.998 ± 0.000	0.994 ± 0.001	0.999 ± 0.000	0.981 ± 0.001
25	0.983	0.982	0.995	0.997 ± 0.000	0.993 ± 0.003	0.998 ± 0.000	0.994 ± 0.001	0.999 ± 0.001	0.981 ± 0.001
30	0.983	0.981	0.995	0.997 ± 0.000	0.993 ± 0.003	0.997 ± 0.000	0.994 ± 0.001	0.998 ± 0.001	0.981 ± 0.001
35	0.983	0.981	0.995	0.997 ± 0.000	0.993 ± 0.003	0.997 ± 0.001	0.994 ± 0.001	0.998 ± 0.000	0.980 ± 0.001
40	0.982	0.981	0.995	0.996 ± 0.000	0.993 ± 0.003	0.996 ± 0.001	0.994 ± 0.001	0.998 ± 0.000	0.980 ± 0.001
45	0.982	0.981	0.995	0.996 ± 0.000	0.992 ± 0.003	0.996 ± 0.001	0.993 ± 0.001	0.998 ± 0.000	0.980 ± 0.001
50	0.982	0.981	0.994	0.996 ± 0.000	0.992 ± 0.003	0.996 ± 0.001	0.993 ± 0.001	0.998 ± 0.000	0.980 ± 0.001

(1) **2CKG4ER achieves superior Adaptability performance.** The proposed 2CKG4ER consistently achieves higher Adaptability values than all baseline methods across the three datasets when $N \leq 20$. Specifically, 2CKG4ER obtains the highest Ada scores across all recommendation sizes on Eedi-sub, with a maximum improvement of approximately 2% over the second-best method, TransE-adv. On Algebra 2005, 2CKG4ER achieves the best Ada performance among all baselines for recommendation sizes $N \geq 5$, outperforming the second-best method by 2.9%. On XES3G5M-sub, 2CKG4ER reaches the highest Ada score of 0.806 when the recommendation size is 10, compared with 0.787 achieved by the second-best method, TransE.

(2) **2CKG4ER is robust to increasing recommendation sizes.** As the recommendation list size N increases from 5 to 50, traditional KGE models experience noticeable performance degradation due to the increasing difficulty of maintaining high-quality recommendations with larger candidate sets. For example, on Algebra 2005, the Ada score of TransE-adv decreases by 20.1% (from 0.731 at $N = 5$ to 0.530 at $N = 50$). In contrast, 2CKG4ER demonstrates stronger robustness, with only a 12.6% decrease (from 0.726 at $N = 5$ to 0.600 at $N = 50$). This resilience can be attributed to the proposed feature representation module, which integrates multi-modal exercise content with cognitive features, such as difficulty and discrimination, thereby providing richer exercise representations for long-tail exercise recommendation.

(3) **KGE-based methods significantly outperform CF-based methods across all settings.** CF-based methods may struggle to capture dynamic cognitive changes during the learning process due to their limited ability to model complex relational information. By explicitly modeling multi-relational interactions through the cognitive attributed knowledge graph $\mathcal{G}$, 2CKG4ER effectively captures the relationships among students, exercises, and knowledge concepts. Furthermore, translational KGE models (e.g., TransE and RotatE) consistently outperform bilinear models (e.g., DistMult and ComplEx), suggesting that translation-based distance modeling is more suitable for preserving relational patterns in educational knowledge graphs.

Table 7

Performance comparison of different methods on Algebra 2005 under varying recommendation sizes N (Mean $\pm$ Std).

(a) Adaptability performance (Ada).

N	EB-CF	SB-CF	CBF	RotatE	DistMult	ComplEx	TransE	TransE-adv	2CKG4ER
5	0.224	0.368	0.415	0.682 ± 0.003	0.424 ± 0.036	0.389 ± 0.030	0.713 ± 0.003	0.731 ± 0.002	0.726 ± 0.002
10	0.221	0.464	0.404	0.655 ± 0.006	0.391 ± 0.026	0.355 ± 0.016	0.705 ± 0.002	0.714 ± 0.004	0.726 ± 0.001
15	0.220	0.466	0.386	0.626 ± 0.012	0.360 ± 0.020	0.331 ± 0.010	0.696 ± 0.002	0.687 ± 0.006	0.715 ± 0.000
20	0.219	0.446	0.376	0.602 ± 0.011	0.340 ± 0.016	0.315 ± 0.008	0.686 ± 0.003	0.660 ± 0.001	0.705 ± 0.006
25	0.218	0.424	0.371	0.578 ± 0.006	0.330 ± 0.013	0.306 ± 0.007	0.654 ± 0.002	0.636 ± 0.008	0.683 ± 0.009
30	0.218	0.402	0.366	0.559 ± 0.001	0.321 ± 0.011	0.298 ± 0.006	0.631 ± 0.003	0.613 ± 0.007	0.656 ± 0.007
35	0.217	0.382	0.362	0.545 ± 0.001	0.313 ± 0.009	0.292 ± 0.004	0.614 ± 0.003	0.595 ± 0.003	0.638 ± 0.007
40	0.216	0.365	0.358	0.529 ± 0.001	0.307 ± 0.008	0.287 ± 0.003	0.600 ± 0.007	0.575 ± 0.002	0.626 ± 0.006
45	0.216	0.350	0.352	0.513 ± 0.002	0.301 ± 0.007	0.282 ± 0.002	0.589 ± 0.007	0.552 ± 0.003	0.615 ± 0.004
50	0.216	0.339	0.346	0.498 ± 0.003	0.297 ± 0.005	0.278 ± 0.001	0.576 ± 0.006	0.530 ± 0.001	0.600 ± 0.002

(b) Novelty performance (NOV).

N	EB-CF	SB-CF	CBF	RotatE	DistMult	ComplEx	TransE	TransE-adv	2CKG4ER
5	0.957	0.975	0.987	0.994 ± 0.000	0.987 ± 0.005	0.986 ± 0.005	0.987 ± 0.001	0.990 ± 0.005	0.985 ± 0.002
10	0.962	0.979	0.988	0.993 ± 0.001	0.986 ± 0.006	0.987 ± 0.004	0.985 ± 0.002	0.988 ± 0.002	0.986 ± 0.002
15	0.964	0.982	0.988	0.991 ± 0.001	0.985 ± 0.005	0.988 ± 0.003	0.985 ± 0.002	0.987 ± 0.002	0.985 ± 0.000
20	0.964	0.981	0.988	0.990 ± 0.001	0.985 ± 0.003	0.988 ± 0.003	0.985 ± 0.001	0.985 ± 0.001	0.985 ± 0.001
25	0.965	0.981	0.988	0.989 ± 0.000	0.985 ± 0.002	0.987 ± 0.002	0.985 ± 0.001	0.985 ± 0.003	0.986 ± 0.001
30	0.966	0.981	0.989	0.989 ± 0.000	0.985 ± 0.002	0.988 ± 0.002	0.984 ± 0.000	0.984 ± 0.003	0.986 ± 0.001
35	0.965	0.980	0.990	0.989 ± 0.000	0.985 ± 0.001	0.988 ± 0.002	0.983 ± 0.000	0.984 ± 0.002	0.986 ± 0.001
40	0.965	0.979	0.990	0.989 ± 0.000	0.985 ± 0.002	0.988 ± 0.002	0.983 ± 0.001	0.983 ± 0.000	0.986 ± 0.001
45	0.965	0.980	0.989	0.989 ± 0.000	0.985 ± 0.002	0.987 ± 0.002	0.983 ± 0.001	0.982 ± 0.000	0.985 ± 0.000
50	0.964	0.980	0.989	0.989 ± 0.000	0.985 ± 0.002	0.987 ± 0.002	0.983 ± 0.001	0.982 ± 0.001	0.985 ± 0.001

(4) **2CKG4ER achieves high Adaptability without compromising Novelty.** Across all three datasets, 2CKG4ER maintains consistently high Novelty scores ($\approx 0.980 - 1.000$), matching or closely approaching the performance of the best-performing baseline methods. This demonstrates that 2CKG4ER effectively balances educational suitability (Ada) and exercise diversity (Nov), while avoiding the tendency to repeatedly recommend highly similar or overly familiar exercises.

5.5.2. Ablation Study (RQ2)

To evaluate the contributions of individual features and the feature representation module (Section 4.3 in 2CKG4ER), we conduct an ablation study on the Eedi-sub dataset across varying recommendation sizes N , as reported in Table 9. “No-student-feature” indicates that student’s ability profile θ is not used in the feature representation. Instead, each student is represented using an ID-based embedding. Similarly, “No-Exercise-feature” denotes that exercises are represented by ID embeddings rather than by semantic content and cognitive parameters. “No-mastery” indicates that the mastery relation is removed from the cognitive knowledge graph. Similarly, “No-forgetting” indicates that the exercise forgetting relation is excluded for the cognitive knowledge graph. “ID-based Only” replaces the continuous feature and relation representations with discrete ID-based embeddings, where all entities and relations are represented using IDs.

The empirical findings reveal three key insights.

(1) **Full feature-based model (our 2CKG4ER) consistently achieves higher Adaptability** across all recommendation sizes N compared with *ID-based Only* variant. This confirms that preserving continuous cognitive attributes and multi-modal entity features via MLP-based fusion provides richer cognitive representations than static ID-based embedding.

(2) **Among relational attributes, student-concept mastery (No-Mastery) proves to be the most critical component.** Removing mastery relations leads to the most severe performance drop across the entire table (dropping to 0.690 –

Table 8

Performance comparison of different methods on XES3G5M-sub under varying recommendation sizes N (Mean $\pm$ Std).

(a) Adaptability performance (Ada).

N	EB-CF	SB-CF	CBF	RotatE	DistMult	ComplEx	TransE	TransE-adv	2CKG4ER
5	0.633	0.683	0.538	0.663 ± 0.010	0.742 ± 0.024	0.733 ± 0.009	0.787 ± 0.015	0.786 ± 0.011	0.805 ± 0.032
10	0.635	0.691	0.544	0.667 ± 0.007	0.734 ± 0.019	0.732 ± 0.009	0.787 ± 0.012	0.784 ± 0.016	0.806 ± 0.026
15	0.646	0.688	0.550	0.671 ± 0.006	0.728 ± 0.016	0.731 ± 0.008	0.784 ± 0.010	0.778 ± 0.015	0.798 ± 0.021
20	0.650	0.688	0.554	0.674 ± 0.004	0.725 ± 0.015	0.731 ± 0.007	0.782 ± 0.009	0.772 ± 0.015	0.786 ± 0.012
25	0.650	0.688	0.557	0.678 ± 0.002	0.722 ± 0.015	0.728 ± 0.005	0.781 ± 0.007	0.768 ± 0.013	0.778 ± 0.007
30	0.649	0.687	0.559	0.682 ± 0.002	0.719 ± 0.014	0.725 ± 0.004	0.778 ± 0.006	0.765 ± 0.010	0.772 ± 0.006
35	0.650	0.686	0.561	0.685 ± 0.002	0.719 ± 0.013	0.723 ± 0.003	0.775 ± 0.005	0.763 ± 0.009	0.768 ± 0.006
40	0.651	0.683	0.563	0.687 ± 0.002	0.720 ± 0.012	0.722 ± 0.004	0.772 ± 0.005	0.761 ± 0.008	0.765 ± 0.005
45	0.653	0.682	0.565	0.689 ± 0.002	0.720 ± 0.011	0.720 ± 0.003	0.770 ± 0.004	0.759 ± 0.007	0.761 ± 0.005
50	0.654	0.680	0.567	0.691 ± 0.001	0.721 ± 0.010	0.719 ± 0.003	0.767 ± 0.003	0.756 ± 0.006	0.759 ± 0.003

(b) Novelty performance (NOV).

N	EB-CF	SB-CF	CBF	RotatE	DistMult	ComplEx	TransE	TransE-adv	2CKG4ER
5	0.998	1.000	1.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	0.999 ± 0.001
10	0.998	1.000	1.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	0.999 ± 0.001
15	0.999	1.000	1.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.001
20	0.999	0.999	1.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	0.999 ± 0.000
25	0.998	0.999	1.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	0.999 ± 0.000
30	0.996	0.998	1.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	0.998 ± 0.000
35	0.994	0.997	1.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	0.998 ± 0.001
40	0.993	0.998	1.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	0.998 ± 0.001
45	0.993	0.998	1.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	0.998 ± 0.000
50	0.992	0.998	1.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	0.998 ± 0.000

0.696), demonstrating that explicit mastery level modeling forms the foundational backbone for personalized difficulty alignment. On the entity level, removing student cognitive ability representations (*No-student-feature*) causes a substantial performance decline across all N (dropping to 0.698–0.705). Surprisingly, student ability parameters exert a stronger influence on recommendation quality than exercise semantic and cognitive features (*No-Exercise-feature*), despite the exercise features containing rich text semantics and item parameters. This indicates that accurate learner state modeling is more decisive for difficulty matching than static exercise content attributes.

(3) *Scale imbalance in recommendation relational weight $W_{i,j}$ ($N \geq 30$)*. Overall, the full 2CKG4ER model yields optimal performance for small-to-medium list sizes ($N \leq 25$). However, when the recommendation list expands to $N \geq 30$, the variant without temporal forgetting (*No-Forgetting*) slightly surpasses the full version (reaching 0.721–0.723). This phenomenon is mathematically attributed to the scale imbalance between the forgetting rate (exfr) and concept mastery (cmkc) during triple weight calculation ($W_{i,j}$, see Equation 7). Statistical analysis reveals that exercise forgetting scores exhibit significantly smaller values and high sparsity compared to mastery levels across datasets (e.g., on Eedi-sub, exfr has a mean of 0.288 versus 0.493 for mlkc; on Algebra 2005, exfr drops to a mean of 0.094; on XES3G5M, exfr has a mean of 0.300 versus 0.501 for mlkc). In Equation 7, when N expands into less-practiced candidate exercises where $\text{cfr} \rightarrow 0$, the numerical contribution of the forgetting distance term $(\delta_f - \text{cfr}_{i,e}^{t+1})^2$ becomes largely uninformative and uniform. This low-magnitude, static term effectively dilutes the variance of the primary cognitive mastery term $(\delta_m - \prod \text{cmkc}_{i,k}^{t+1})^2$ during positive triple construction. Consequently, omitting the forgetting component eliminates this masking effect, allowing the model to rank deep candidate items purely based on unadulterated mastery calibration. This findings suit sparse datasets when recommendation sizes are large.

Table 9

Ablation study on Eedi-sub under varying recommendation sizes N for Adaptability (Ada) performance (Mean $\pm$ Std).

N	No-student-feature	No-Exercise-feature	No-Mastery	No-Forgetting	ID-based Only	2CKG4ER
5	0.705 $\pm$ 0.003	0.722 $\pm$ 0.002	0.696 $\pm$ 0.003	0.713 $\pm$ 0.004	0.713 $\pm$ 0.007	0.722 $\pm$ 0.006
10	0.705 $\pm$ 0.002	0.724 $\pm$ 0.003	0.691 $\pm$ 0.002	0.717 $\pm$ 0.002	0.713 $\pm$ 0.009	0.723 $\pm$ 0.001
15	0.703 $\pm$ 0.001	0.722 $\pm$ 0.002	0.690 $\pm$ 0.001	0.718 $\pm$ 0.003	0.712 $\pm$ 0.011	0.723 $\pm$ 0.003
20	0.702 $\pm$ 0.001	0.720 $\pm$ 0.003	0.691 $\pm$ 0.001	0.720 $\pm$ 0.002	0.711 $\pm$ 0.011	0.722 $\pm$ 0.004
25	0.701 $\pm$ 0.000	0.719 $\pm$ 0.003	0.693 $\pm$ 0.001	0.721 $\pm$ 0.002	0.710 $\pm$ 0.011	0.722 $\pm$ 0.003
30	0.700 $\pm$ 0.000	0.718 $\pm$ 0.003	0.692 $\pm$ 0.002	0.722 $\pm$ 0.001	0.710 $\pm$ 0.012	0.721 $\pm$ 0.002
35	0.699 $\pm$ 0.001	0.717 $\pm$ 0.002	0.692 $\pm$ 0.002	0.723 $\pm$ 0.001	0.710 $\pm$ 0.011	0.720 $\pm$ 0.001
40	0.699 $\pm$ 0.001	0.717 $\pm$ 0.001	0.692 $\pm$ 0.001	0.722 $\pm$ 0.001	0.709 $\pm$ 0.010	0.720 $\pm$ 0.002
45	0.698 $\pm$ 0.001	0.716 $\pm$ 0.001	0.691 $\pm$ 0.001	0.722 $\pm$ 0.001	0.708 $\pm$ 0.009	0.719 $\pm$ 0.001
50	0.698 $\pm$ 0.001	0.716 $\pm$ 0.001	0.692 $\pm$ 0.001	0.721 $\pm$ 0.001	0.708 $\pm$ 0.009	0.718 $\pm$ 0.001

5.5.3. Interpretable Evidence (RQ3)

In order to demonstrate the interpretability of the proposed 2CKG4ER, we utilized the quantitative (questionnaires) method to assess the helpfulness and reasonableness of recommendation reasons in exercise selection.

Questionnaires: Inspired by (Guan et al., 2025), we designed three questionnaires (see <https://github.com/Tangfgjk/KG4ER>). Each questionnaire depicts a student's ability profile, the recommended exercise, and the recommendation reasons. In particular, the recommendation reasons comes from the following four factors: namely mastery cmkc, forgetting cfr, exercise difficulty diff, and exercise discrimination disc, which are the extracted features in Section 4.1.

- Mastery: representing a student's master level over knowledge concepts,
- Forgetting: showing a student's forgetting rate on an exercise,
- difficulty, representing the difficulty level of an exercise, which shows whether the recommended exercises match a student's mastery level,
- discrimination, representing whether the exercise can distinguish students' ability.

The questionnaire assess recommendation reasons from two aspects: namely helpfulness and reasonableness. Each aspect contains four inquiry questions, which correspond to the four factors of the recommendation reason. Therefore, each questionnaire contains eight inquiry questions.

- Helpfulness: Participants evaluated the helpfulness of the recommendation reasons (mastery, forgetting, difficulty, and discrimination) in guiding their exercise selection using a five-point Likert scale, where 1 denotes no assistance and 5 denotes substantial assistance.
- Reasonableness: Participants assessed the logic of the recommendation reasons by rating their reasonableness on a five-point Likert scale, where 1 denotes the least reasonable and 5 denotes the most reasonable.

Participants: A total of 35 post-graduate students were involved to the three questionnaires via social networks (i.e., QQ from Tencent). Among the participants, 18 of them responded to questionnaire I, and 17 of them responded to questionnaires II-III, respectively.

Results analysis: Since the three questionnaires presented different student ability profiles, recommended exercises, and corresponding recommendation reasons, we first investigated whether variations in student profiles affected participants' evaluations. Specifically, the Kruskal-Wallis H test was employed to compare the scores of the eight questionnaire items among the three questionnaires. The results (see Table 10) demonstrated that none of the questionnaire items exhibited significant differences across the three questionnaires ($p > 0.05$), with the minimum p-value being 0.112. These findings indicate that the differences in student ability profiles did not introduce significant evaluation bias. Consequently, the responses from all three questionnaires were aggregated for subsequent analysis, resulting in 52 valid responses in total.

Table 10

Kruskal–Wallis H test results across three questionnaires.

Dimension	Question	Median Rating			<i>H</i> -statistic	<i>p</i> -value
		Quest. I	Quest. II	Quest. III		
Helpfulness	Q1	4	4.5	4	0.333	0.847
	Q2	5	5	4	4.380	0.112
	Q3	4	5	4	1.738	0.419
	Q4	5	5	4	2.332	0.312
Reasonableness	Q5	5	5	5	1.020	0.600
	Q6	5	5	4	3.671	0.160
	Q7	4	5	4	1.941	0.379
	Q8	5	5	4	0.746	0.689

Table 11

Participant evaluation on the helpfulness and reasonableness of recommendation reasons.

Reason	Helpfulness (%)							Reasonableness (%)						
	5	4	3	2	1	Mean		5	4	3	2	1	Mean	
Mastery over a knowledge concept	48.1	26.9	19.2	3.8	1.9	4.15		59.6	19.2	19.2	1.9	0.0	4.37	
Forgetting rate over an exercise	51.9	32.7	9.6	3.8	1.9	4.29		50.0	25.0	21.2	1.9	1.9	4.19	
Exercise difficulty	38.5	34.6	21.2	5.8	0.0	4.06		48.1	30.8	15.4	1.9	3.8	4.17	
Exercise discrimination	53.8	36.5	9.6	0.0	0.0	4.44		53.8	28.8	15.4	1.9	0.0	4.35	

Note: Based on 52 collected questionnaires evaluated on a 5-point Likert scale.

Helpfulness: 5 = Yes, very much; 4 = Yes, quite much; 3 = Yes, somewhat; 2 = No; 1 = Not at all.

Reasonableness: 5 = Very reasonable; 4 = Quite reasonable; 3 = Reasonable; 2 = Unreasonable; 1 = Very unreasonable.

Table 11 shows the score distribution and mean values of the four recommendation reasons. Overall, scores of all four factors are greater than 4 (5 indicating the full points), representing that participants generally held positive opinions for recommendation reasons (mastery, forgetting, difficulty, and discrimination). Specifically, forgetting and exercise discrimination received the highest scores regarding helpfulness, whereas mastery and exercise difficulty are most highly rated factors for reasonableness. This finding may be attributed to the different roles of these factors in exercise recommendation. Forgetting rate is directly related to the intuitive need for review, making its helpfulness easier for users to perceive. Exercise discrimination helps users understand whether an exercise can accurately evaluate their current ability, leading to high evaluations in both helpfulness and reasonableness. Although mastery was considered a highly reasonable factor (59.6% selecting “very reasonable”), its perceived helpfulness was relatively lower (48.1% selecting “very helpful”), possibly because users were more concerned with identifying what to practice next rather than understanding what they had already mastered. Moreover, Friedman test did not reveal significant differences among the four types of recommendation reasons. The helpfulness dimension showed a marginal trend toward significance ($\chi^2(3) = 7.132, p = 0.068$), while the reasonableness dimension remained non-significant ($\chi^2(3) = 3.554, p = 0.314$).

In summary, these findings suggest that participants were generally satisfied with the recommendation reasons provided by the proposed 2CKG4ER. The consistently high evaluation scores indicate that the generated recommendation reasons are understandable and effectively reveal the factors influencing exercise recommendations, demonstrating the interpretability of the proposed 2CKG4ER.

5.5.4. A Case Study

To further illustrate the interpretability of our proposed 2CKG4ER, we randomly selected a student (s_{285}) from the Eedi-sub dataset. Figure 3 compares the recommended exercises generated by SB-CF, TransE-adv, and our 2CKG4ER. SB-CF recommends exercise (e_8) for student (s_{285}) based on a similarity score of 0.56 with previously practiced exercises. However, the recommendation explanation only considers the similarity dimension, which provides limited

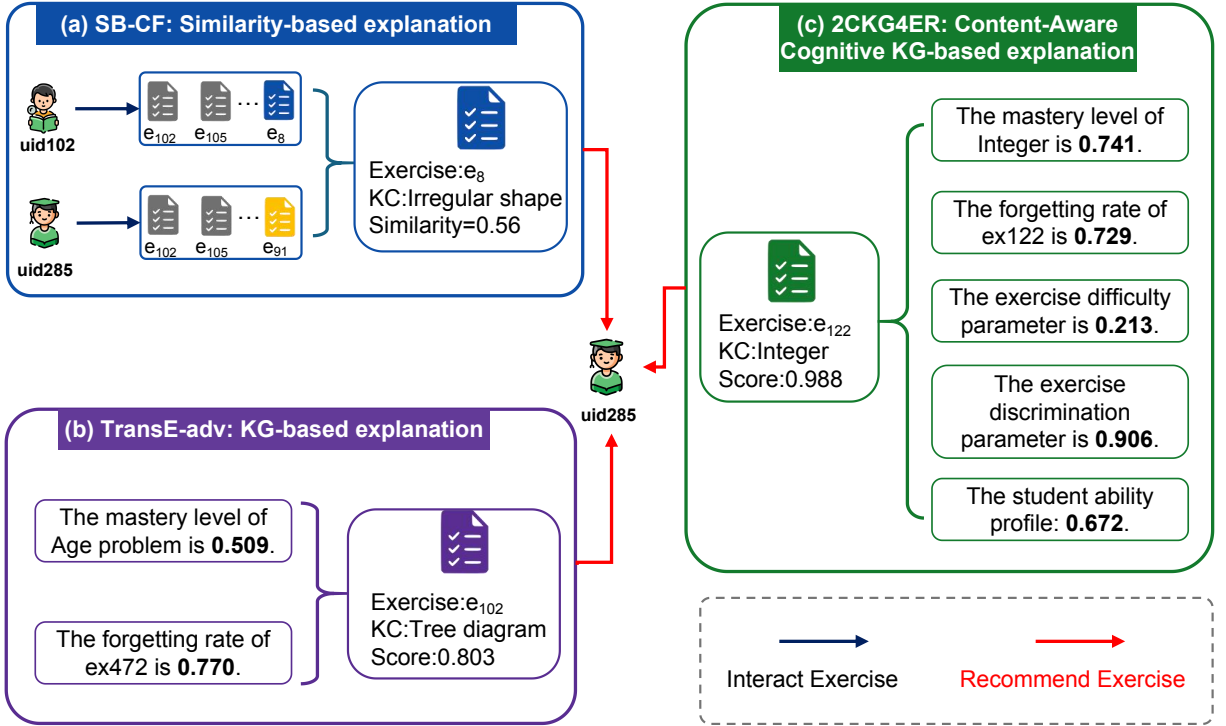


Figure 3: A case study: interpretable comparison among SB-CF, TransE-adv, and our 2CKG4ER.

interpretability. TransE-adv recommends exercise (e_{102}) based on the mastery level (0.509) and forgetting rate (0.770); however, it does not provide explanations related to exercise features. In contrast, 2CKG4ER recommends exercise (e_{122}) with a mastery level of 0.741 and a forgetting rate of 0.729. Furthermore, the recommended exercise has a low difficulty parameter of 0.213, which is lower than the student's mastery level (0.741), indicating that the exercise presents an appropriate challenge for (s_{285}). Recommending exercises that require slightly higher abilities aligns with the concept of the zone of proximal development, which can facilitate students' ability improvement. In addition, the high exercise discrimination value (0.906) suggests that the recommended exercise can effectively assess (s_{285})'s ability, given the student's relatively high overall ability profile (0.672).

In summary, 2CKG4ER demonstrates superior interpretability compared with SB-CF and TransE-adv. Specifically, 2CKG4ER not only provides personalized exercise recommendation but also generates comprehensive explanations by incorporating both relational information and the intrinsic features of students and exercises.

6. Conclusion

In this study, we proposed 2CKG4ER, interpretable personalized exercise recommendation method based on a content-aware cognitive knowledge graph. By incorporating shared semantic and cognitive exercise features across both student modeling (ECAKT) and exercise recommendation, we bridge two traditionally separate tasks to enhance recommendation performance, directly addressing the limitation of conventional models that fail to propagate exercise properties from knowledge tracing to recommendation. Based on the extracted features of entities and relationships, we constructed an attributed cognitive knowledge graph. We then adopted an MLP operator to fuse features from multiple sources and devised the ReConvE knowledge graph embedding model to learn the interactions between students and exercises. Experimental results demonstrate that 2CKG4ER achieved superior performance across three datasets compared to eight baselines. While ReConvE operates as a latent embedding-based model, the proposed 2CKG4ER provides clear recommendation rationales by using rich entity and relational features in the attributed cognitive knowledge graph.

Although 2CKG4ER achieves strong performance, it has a few limitations. First, our feature representation relies on basic concatenation and MLP operators, which could be upgraded to advanced attention-based or hypergraph fusion mechanisms. Second, student profiles are currently modeled using only the cognitive ability parameter, leaving room to incorporate multi-dimensional learner behaviors. In the future, we plan to enhance the feature fusion architecture and enrich learner profiles to further improve the performance of 2CKG4ER.

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